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vdcmaniac

A modern Commodore 128 VDC (8563/8568, 80-column chip) demo, written from scratch in C with the Oscar64 cross-compiler.

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- Further documentation: `ARCHITECTURE.md` (code layout), `vdc_reference_manual.md` (VDC registers/modes), `raster_lib_manual.md` (raster bar/IRQ library), `krill_manual.md` (fastloader integration)

Inspiration

This project is inspired by **VDC Mode Mania**, a 2012 demo by **Tokra** and **Mike** of **Akronyme Analogiker**, originally released at the Connected 9 party (Sep 2, 2012), later extended twice (an 800x600 mode in 2015, a 960x540 mode in 2023). VDC Mode Mania was a slideshow showcasing the C128's rare, little-used 64KB-VDC-RAM graphics modes -- interlaced and non-interlaced colour and monochrome modes at resolutions far beyond the VIC-II's usual 320x200, from 480x252 colour up to 960x540 monochrome -- with converted picture assets and BASIC-based viewers. The original demo, including its BASIC source and image converters, is available from CSDb: https://csdb.dk/release/?id=234174

A second main inspiration: Peter Hulstede's "VDC-Intromaker: Perfektes Rasterzeilen-Timing", **64'er Sonderheft 95**, p.45 -- its example CIA-timer VDC raster-sync routine (SYNC/WAITLINE/WAITJUMP/LINEN 6502 listing) is the basis this project's own `raster_synch()`/`raster_waitline()` (`include/vdc_raster.c`) are translated from into C, driving every raster-bar effect in the demo -- see that file's own credit comment for the full technical mapping.

`vdcmaniac` is **not a port** of that BASIC codebase. It's a from-scratch reimplementation in C, targeting the same family of rare VDC modes but built around this project's own tooling and techniques:

- **Oscar64**, a modern C cross-compiler for 6502 targets, instead of interpreted BASIC.

- **Krill's fastloader**, with on-the-fly TSCrunch decompression, for asset loading -- see `krill_manual.md`.
- A **custom raster-IRQ trampoline**, installed directly on the CPU's hardware interrupt vector (not the KERNAL soft vector, which this project's own boot sequence makes unreliable), driving per-scanline VDC colour effects and, going forward, synced SID music playback -- see `raster_lib_manual.md`.
- A **from-scratch VDC mode/register library** (`include/vdc_core.c`) and its own picture converters (`tools/vdc_convert.py`) rather than the original's PC-side C converters -- see `vdc_reference_manual.md`.

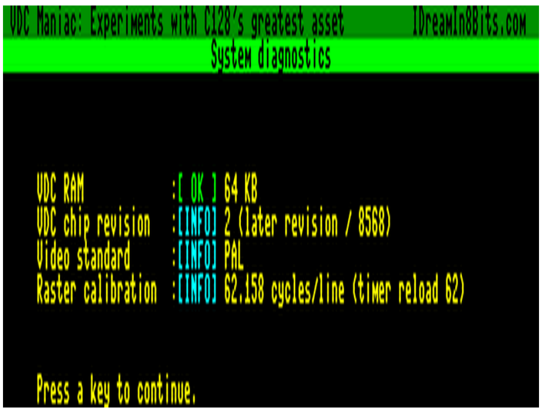
The set of VDC modes being brought back is the same one VDC Mode Mania demonstrated (see the original's own README, linked above, for the full list and Tokra's own notes on real-hardware/monitor compatibility per mode); the techniques driving them are new.

On pictures: every showcase section uses this project's own sourced artwork, converted via `tools/vdc_convert.py` -- public-domain paintings/ prints and CC-licensed photographs, each with a full attribution/licence note in `include/defines.h`'s own credit block. None of Tokra's original converted images are used.

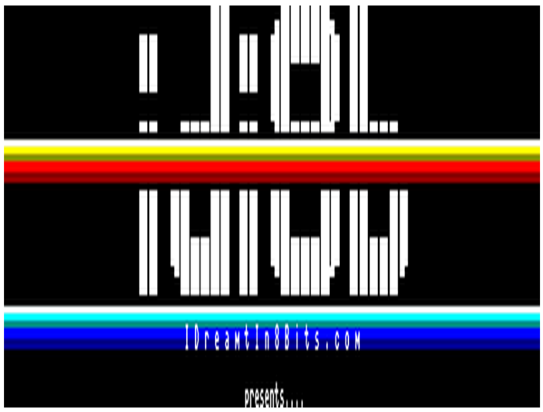
Screenshots

Captured in VICE (x128). VDC pixels are not square, so a raw capture misrepresents every mode's true on-screen proportions -- each screenshot below is cropped to that mode's own visible content and restored to a standard 4:3 aspect ratio. Full picture licence/attribution detail lives in `include/defines.h`'s own credit block; the captions below summarise it per picture.

Intro sequence



System diagnostics (PAL/NTSC, VDC revision, 64KB check)

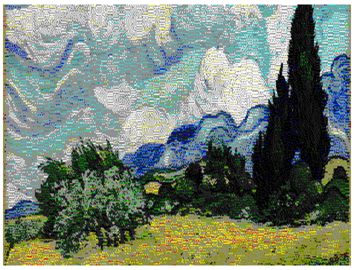


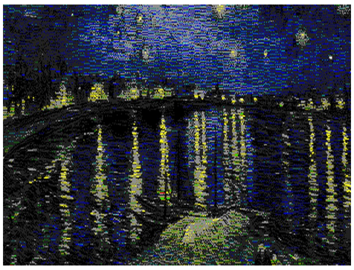
idi8b logo section

Title screen, raster bar underneath

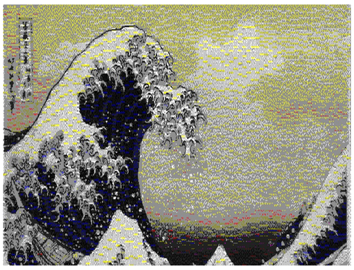
Main menu, raster-bar row highlight

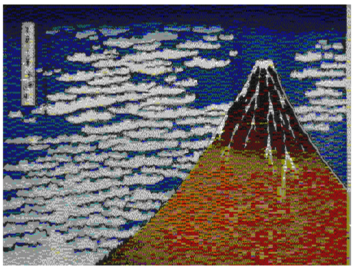
VDC-FLI (480x252, colour, non-interlace) + VDC-HFLI (640x400, colour, non-interlace)


”Wheat Field with Cypresses” (1889), Vincent van Gogh, The Met, public domain


”Starry Night Over the Rhone” (1888), Vincent van Gogh, Musee d'Orsay, public domain


”Vase with Irises Against a Yellow Background” (1890), Vincent van Gogh, Van Gogh Museum, public domain


”The Great Wave off Kanagawa” (c. 1831), Katsushika Hokusai, public domain


”Fine Wind, Clear Morning” (”Red Fuji”, c. 1830-1832), Katsushika Hokusai, public domain

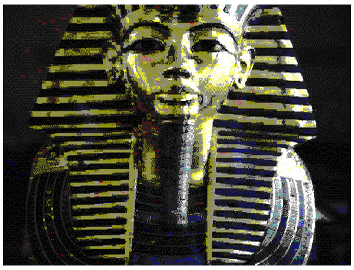

”Ejiri in Suruga Province” (c. 1830-1832), Katsushika Hokusai, public domain

VDC-IFLI: VDC-IHFLI (640x480, colour, interlace) + VDC-ITFLI (640x576, colour, interlace)

”Passiflora caerulea (makro close-up)” by Petar Milosevic, CC BY-SA 4.0

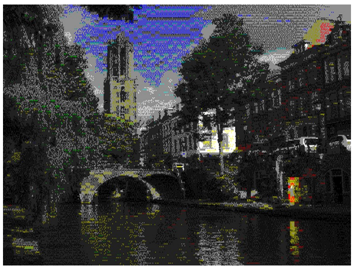
”Sonnenblume Helianthus 1” by Bohringer Friedrich, CC BY-SA 2.5

”Keel-billed Toucan, Caves Branch Jungle Lodge, Belize” by Judy Gallagher, CC BY 2.0



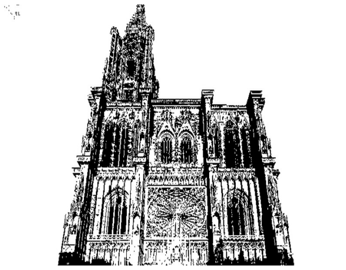
”Tutanchamun Maske” (funerary mask) by MykReeve, CC BY-SA 3.0 / GFDL 1.2+

”Hyacinth macaw head”, the Pantanal, Brazil, by Charles J. Sharp, CC BY-SA 4.0

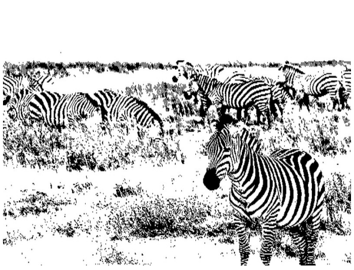


”De Hamburgerbrug met de Oudegracht en de Domtoren in de Stad Utrecht” by Jan dijkstra, CC BY-SA 4.0

VDC-mono: VDC-IMONO (720x700, interlace) + VDC-IM800 (800x600, interlace)

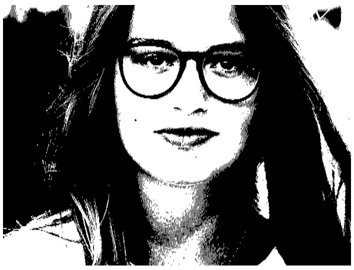


”Strasbourg Cathedral Exterior” by David Iliff (Diliff), CC BY-SA 3.0 / GFDL 1.2+



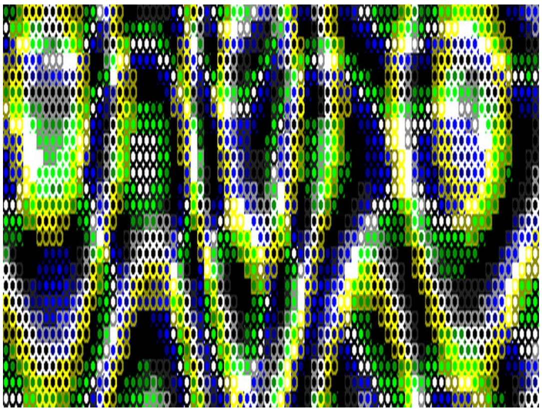
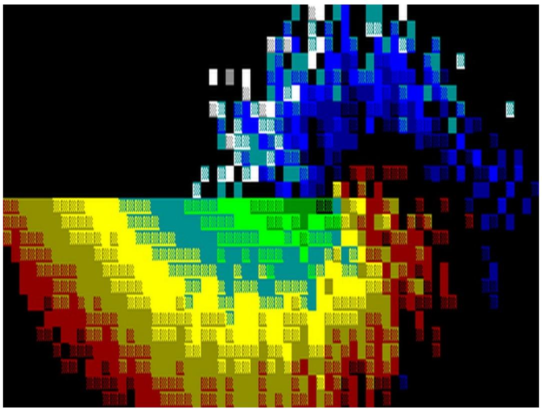
”Zebras Ngorongoro Crater” by Muhammad Mahdi Karim, GFDL 1.2

”Portrait de femme en tenue traditionnelle de Berbere Algerien” by Samia Dib Benkaci, CC BY-SA 4.0

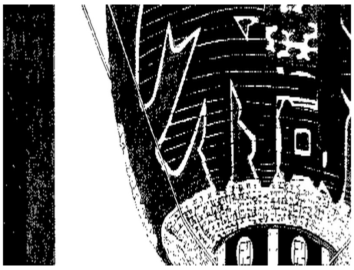
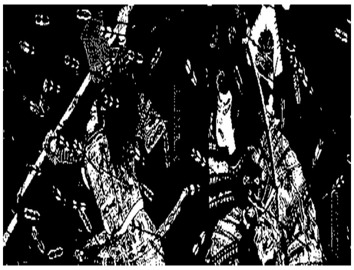


”Portrait-of-a-woman” by Mark Sherman, CC BY 2.0	”The History of Apple Pie - Kelly Lee Owens (2013)” by Sylvain lasco, CC BY-SA 4.0	”Maupi”, the author's own cat

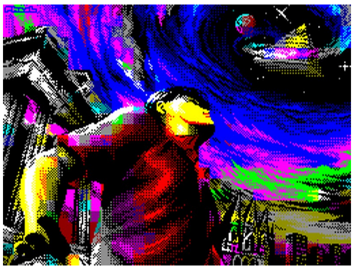
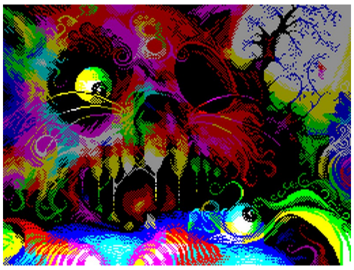
Plasma effect / Colour rotation effect

	
Sine-table plasma, own procedural content	Colour-cycling bitmap, own procedural content

VDC-SCROLL: VDC-VSCROLL + VDC-PANORAMA + VDC-PANORAMA 2D

		
”Kinryuzan Temple, Asakusa”, print #99 from One Hundred Famous Views of Edo (1856), Utagawa Hiroshige, public domain -- vertical scroll	”Nine Dragons” (1244), Chen Rong, Museum of Fine Arts, Boston, public domain -- horizontal pan	”The Last Stand of the Kusunoki at Shijonawate” (1857), Utagawa Kuniyoshi, British Museum, public domain -- combined 2D pan

VDC Spectrum (256x192, doubled to 512x192)

		
”np” (2015), prof4d, 1st place at CC Winter - DiHalt Lite 2015	”Prisoner of Time” (2001), PheeL, 1st place at Chaos Constructions 2001	”Cursed Eighth” (2010), Piesiu, 1st place at the Chaos Constructions 2010 ZX Graphics compo

End demo + credits

Scrolling-text and colour-cycling-bars credits sequence.

Known issues

- **Monitor compatibility varies by VDC mode.** Not every one of the VDC's rare bitmap modes displays cleanly on every monitor -- this is a real-hardware VDC characteristic, not a bug in this demo (see the original VDC Mode Mania's own README, linked in ”Inspiration” above, for Tokra's own notes on the same thing). A **Commodore 1901** monitor is confirmed to show every mode correctly, but may still need its own horizontal/vertical size and centering controls adjusted per mode -- as may any other monitor.
- **z64k**: the title screen renders mangled. Every other mode is unaffected. VICE shows every mode correctly.
- **RGBtoHDMI**: each VDC mode needs its own RGBtoHDMI mode definition. This project does not currently provide RGBtoHDMI mode definitions for any of its VDC modes.
- **VICE/z64k under WSL2/WSLg**: SID music playback can noticeably slow down in some sections -- a WSL2/WSLg performance characteristic, not reproduced on native Windows/Linux or real hardware. For accurate timing/audio, prefer running VICE/z64k natively rather than under

WSL2.

Requirements

- **Krill's fastloader requires exclusive use of the IEC bus.** Only one device may be active on the bus: the drive holding the demo disk, at device ID 8. Power off (or otherwise disable) any other IEC device -- printers, second drives, etc. -- before running the demo.
- **Only a D81 disk image is supported.** D71 and especially D64 would need far more disk swapping to fit this project's own asset sizes, so neither is built or supported.
- **SD2IEC is not supported.** Krill's loader needs cycle-exact IEC bus timing, which SD2IEC's own IEC emulation does not provide. Use a real 1541/1571/1581 (or equivalent, e.g. an Ultimate II+) drive.

Building from source

Prerequisites

Tool	Purpose	Install
oscar64	C compiler targeting C128	Build from source or download release
`c1541`	Disk image creation (part of VICE)	`sudo apt install vice`
`wput`	FTP upload to Ultimate II+	`sudo apt install wput`
`java` (JRE 8+)	Run z64k (`make z64k`)	`sudo apt install default-jre` (optional)
`pandoc` + `texlive-xetex`	Regenerate README.pdf from README.md	`sudo apt install pandoc texlive-xetex` (optional)
`python3` + Pillow	Regenerate converted picture assets	`pip install Pillow` (optional -- build warns and skips if missing)

The compiler is expected at `/home/xahmol/oscar64/bin/oscar64`. Edit the `CC` variable in the Makefile if yours is installed elsewhere.

Deployment setup (.env)

Create a `.env` file in the project root to configure deployment to your Ultimate II+. This file is gitignored and will never be committed:

```
ULTIP1 = 192.168.1.xx       # IP of your primary Ultimate II+
# ULTIP2 = 192.168.1.yy    # optional second machine
# ULTIP3 = 192.168.1.zz    # optional real-hardware test machine (make deploy3)
```

The Makefile constructs the FTP path for `deploy`/`deploy2` as `ftp://$(ULTIP1)/usb1/temp/`. Override `ULTUSB ?= usb1` in `.env` if your USB port is numbered differently. `deploy3` uses its own dedicated path (`ULTPATH3`, default `/USB1/idi8b/dev/`) instead -- override that in `.env` too if you want it elsewhere.

Make targets

Target	Description
`make` / `make all`	Build the program, boot sector and the Krill-fastloader D81 disk image (see `krill_manual.md`)
`make krill`	Same D81 build, invokable on its own
`make clean`	Remove all build artefacts
`make vice`	Launch VICE x128 with the D81 -- **needs True Drive Emulation enabled** (VICE Settings -> Drive, or `x128 -drive8truedrive`); Krill's loader runs drive-side install code (`M-E`/`M-R`) that silently hangs at the "loading assets" screen without it
`make z64k`	Launch z64k with the D81 (downloads `Z64K.jar` on first use; Java 8+ required) -- a second, independently-implemented C128/VDC emulator, useful when VICE doesn't reproduce a real-hardware issue
`make deploy`	Upload build to primary Ultimate II+ (requires `ULTIP1` in `.env`)
`make deploy2`	Upload to second Ultimate II+ (requires `ULTIP2` in `.env`)
`make deploy3`	Upload build to a real-hardware test machine (requires `ULTIP3` in `.env`)
`make docs`	Regenerate `README.pdf` from `README.md` (requires pandoc)

License

This project's own code is licensed under the GNU General Public License v3.0. Krill's fastloader (`krill/`) and TSCrunch (`krill/loader/tools/tscrunch/`) are vendored third-party components under their own, separate, permissive attribution-required licenses -- both are credited in-demo (the end-credits scroller) and in this README's own Inspiration section, per their own license terms. Every picture/artwork/music asset has its own individual attribution and licence noted in `include/defines.h`'s credit block -- see "On pictures" above.